Document Comparison: JSTARS-2025-00807 Minor Revision (Round 2)

Changes to writeup.md since pre-revision version (commit 2d83ecd)

Summary of Changes

```
writeup/writeup.md | 51 ++++++++++++++++++++++++++++++++++++++++++---------
1 file changed, 42 insertions(+), 9 deletions(-)
writeup/writeup.md | 24 ++++++++++++------------
1 file changed, 12 insertions(+), 12 deletions(-)
```

Legend

- ~~Red strikethrough text~~ = Deleted from original
- Green text = Added in revision
- **Blue text** = Line location markers

Detailed Changes

```
diff --git a/writeup/writeup.md b/writeup/writeup.md
index b133ae8..72c4f90 100644
--- a/writeup/writeup.md
+++ b/writeup/writeup.md
@@ -112,7 +112,10 @@ The strength of traditional machine learning approaches lies in their
ability to

 Traditional machine learning algorithms require the extraction of variables (e.g. max EVI,
mean blue band) that can help distinguish different plant or crop types [@begue2018remote].
The development of salient time-series features to capture phenological differences between
locations from remotely sensed images remains a challenge. These features are typically
derived from the spectral bands (e.g. red edge, NIR) of the satellite imagery or indexes, such
as the enhanced vegetation index (EVI), and basic time series statistics (e.g. mean, max,
minimum, slope) for the growing season [@morton2006cropland]. Meanwhile a broader set of time
series statistics from bands or indexes may be more relevant for a number of applications. For
instance the skewness of EVI might help distinguish crops that green-up earlier vs later in
the season, measures of the numbers of peaks in EVI might help differentiate intercropping or
multiple plantings in a season [@begue2018remote]. However, the selection and extraction of
these features can be time-consuming and labor-intensive, requiring domain expertise and
manual intervention.

-In contrast, deep learning methods have dominated the most recent literature
[@agriculture13050965; @hohl2024recent]. These methods include both recurrent neural networks
(RNN) and convolutional neural networks (CNN). Recurrent Neural Networks (RNNs) are a class of
neural networks that are particularly powerful for modeling sequential data such as time
series, speech, text, and audio. The fundamental feature of RNNs is their ability to maintain
a 'memory' of previous inputs by using their internal state (hidden layers), which allows them
to exhibit dynamic temporal behavior. RNNs and its variants allow the integration of time-
series imagery, substantially improving crop type classification outcomes especially in data
rich environments [@agriculture13050965; camps2021deep]. Deep learning approaches however
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~~typically require much larger sets of training data, may be more prone to overfitting especially with small sample sizes, have substantial limitations to interpretability, and require expensive compute [@hohl2024recent; @rs13132591; @agriculture13050965; @LI2023103345; @MA2019166]. Although recent efforts have closed the gap e.g. [@tseng2021cropharvest], the lack of readily available and reliable ground truth data or benchmark datasets for training, as discussed earlier, may limit the applicability of deep learning for a variety of tasks including crop classification and make researchers more reliant of less reliable techniques like transfer learning or zero-shot or low shot methods [@owusu2024towards; @LI2023103345; @MA2019166]. Moreover, training data for extreme events, like crop losses, disease, and lodging are largely non-existant. Interpretability is also a salient weakness as interpretation of models allows us to gain scientific insight and assess trustworthiness and fairness in so far as outputs affect policy decisions.~~

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+of convolutional recurrent neural networks (ConvRNN) on Sentinel-1 synthetic aperature radar (SAR) features and Sentinel-2 surface reflectance. In controlled agricultural settings, like Bradford Research Cetner in Missouri, combining SAR and optical data for similar models are able to achieve much higher overal accuracy levels of 94.1% for 3D U-Net, 84.7 for 2D U-Net, and 64% for SegNet [@ADRIAN2021215]. In Moreover, training data for extreme events, like crop losses, disease, and lodging are largely non-existant. Interpretability is also a salient weakness as interpretation of models allows us to gain scientific insight and assess trustworthiness and fairness in so far as outputs affect policy decisions.

+

+Recent advances in lightweight transfer learning architectures, such as MobileNetV2 and EfficientNet, have shown promising results in agricultural applications, particularly for RGB image classification on mobile devices and smart farming systems [@sandler2018mobilenetv2; @tan2019efficientnet]. These architectures offer reduced computational requirements compared to larger deep learning models while maintaining competitive performance. However, transfer learning approaches face domain adaptation challenges when applied to multispectral satellite imagery, as pre-trained weights are typically derived from natural RGB images rather than remote sensing data. Our feature-based approach offers complementary benefits for satellite-based crop classification: improved interpretability through SHAP values that provide agronomically meaningful explanations, accessibility on standard hardware without GPU requirements, and transparency in the feature engineering process that allows domain experts to incorporate their knowledge directly.

An alternative approach turns back the clock on deep learning approaches. For instance CNN classifiers, through the exertion of tremendous effort of GPUs, can apply and learn from thousands of filters or convolutions that help detect distinct features like edges, textures or patterns. It is however possible to apply a more limited yet salient set of filters like Fourier Transforms, Differential Morphological Profiles [@pesaresi2001new], Line Support Regions or Structural Feature Sets [@huang2007classification] amongst others, to images and then use these as features in more traditional machine learning approaches [@graesser2012image, @owusu2024towards, @engstrom2022poverty; @chao2021evaluating; @urbansci7040116]. This approach may be particularly useful in data-scarce environments, requiring less training data and potentially offering more efficient results in low-information settings. The same approach has been taken for time series analysis, where instead of learning patterns through a RNNs memory, we can apply a more limited but potentially salient series of time series filters. Measures of trends, descriptions of distributions, or measures of change and complexity might adequately describe time series properties for regression and classification tasks [@christ2018time; @yang2021anomaly]. This time series filter approach, developed for this paper, can also be applied on a pixel-by-pixel basis to satellite image bands or index values[@xr_fresh_2021].

@@ -191,7 +194,7 @@ Young crops exhibit substantial differences compared to mature crops in terms of
    \label{fig:crop_cal} %can refer to in text with \ref{fig:crop_cal}
 \end{figure} Source: [@cc_tanzania]

~~-USDA's Foreign Agricultural Service compiles information on planting and harvest windows for grain, oilseed, and cotton crops as an important tool to support crop condition assessments with satellite imagery. Tanzania's crop planting seasons are shaped by its bimodal and unimodal rainfall patterns, which vary by region. In the north and northeast, bimodal areas experience the short rains (Vuli) from late-October to mid-January, during which crops like maize, beans, and vegetables are planted in October and November, and the long rains (Masika) from March to May, supporting crops like maize, rice, sorghum, and cassava, typically planted in February and March. In the central, southern, and western regions with unimodal rainfall, there is a single rainy season from November to April, when crops such as cotton, maize, millet, rice, and sunflower are planted in November and December. This diversity in rainfall patterns allows for a wide variety of crops suited to the local climate and seasonal conditions.~~
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Data collection took place between late April and May 2023 (Figure \ref{fig:crop_cal}) to align with the mid-season growth stage for most target crops. YouthMappers were trained to focus on crops known to be present in each region and at appropriate phenological stages for spectral discrimination. Prior to fieldwork, extensive training sessions covered the key criteria for selecting suitable field sites to ensure high-quality training data for the machine learning models.

@@ -222,10 +225,12 @@ Dodoma & 800 & Sorghum, Maize, Millet, Sunflower, Groundnut, Cotton \\
 \label{tab:data_n}
 \end{table}

~~-### Field Data Cleaning~~
+### Field Data Cleaning and Validation

The initial dataset comprised 1,720 observations collected by YouthMappers across the three districts. A thorough data cleaning process was undertaken to ensure the quality and reliability of the dataset for model training and evaluation. This process involved several steps. First, duplicate entries were identified and removed to prevent redundancy and potential bias in the dataset. Second, observations with missing or incomplete data were addressed; depending on the extent of missing information, these entries were either corrected using auxiliary data sources or excluded from the dataset. Third, each observation was visually inspected using in-situ photos taken by students at each site, which helped verify the accuracy of the recorded crop types and field conditions. Fourth, crop type labels were standardized to ensure consistency across the dataset by correcting typographical errors and unifying different naming conventions for the same crop. Finally, the geographic coordinates of each observation were validated to ensure they fell within the expected study area and corresponded to an observable field from satellite imagery. After completing the cleaning process, the final dataset consisted of over 1,400 crop type entries, providing a robust foundation for training and evaluating the machine learning models used in this study.

+The photo-based validation process served as a critical quality control mechanism. Each field photograph was reviewed by researchers familiar with regional crop characteristics to confirm or correct the crop type labels recorded during field visits. This remote verification allowed identification of misclassified observations where field conditions or crop similarity led to labeling errors. Approximately 320 observations were removed during the cleaning process due

to factors including unclear photographs, label inconsistencies, coordinate errors, or fields that could not be confidently verified. While this photo-based approach provided valuable validation, it represents a limitation of our methodology: independent post-classification field visits to verify model predictions were not conducted. Future studies would benefit from systematic ground truthing campaigns following model application to assess real-world classification accuracy and identify systematic errors in specific crop types or geographic areas.
+
 ## Analytical Methods

 After data collection and cleaning, we employed a series of analytical methods to extract relevant features from the satellite imagery, train machine learning models, and evaluate their performance. The overall workflow is illustrated in Figure \ref{fig:analyt_flow}.
@@ -273,19 +278,43 @@ The final model selection was based on maximizing the kappa statistic across all

 To assess model performance, we implemented stratified group k-fold cross-validation with three splits. This approach ensured that samples from the same field remained together within either the training or validation set, preventing data leakage that could occur if observations from the same field were split across folds. We utilized the kappa statistic as our primary evaluation metric due to its suitability for assessing classifier performance on imbalanced datasets. This metric accounts for agreement occurring by chance, providing a more robust measure of classification accuracy than simple overall accuracy when class distributions are unequal.

+### Computational Efficiency
+ A primary advantage of our "lite learning" approach is its computational efficiency, particularly when processing high-dimensional satellite time series. LightGBM optimizes the training process using a histogram-based approach, with a time complexity of:
+
+ $$T_{LGBM} = O(n \cdot m) + O(b \cdot k \cdot d)$$
+
+ Here, the first term represents the one-time cost of binning n samples across m features (where $m = bands \times timestamps$). The second term captures tree construction across b bins, k trees, and maximum depth d. Because split-finding operates on compressed bins rather than raw data points, the expensive $O(n \cdot m)$ operation occurs only once. In our implementation, hyperparameter optimization (100 Optuna trials, 3-fold CV) concluded within 10–15 minutes on a standard laptop (Intel i7, 16GB RAM) without GPU acceleration.
+
+ This efficiency stands in stark contrast to deep learning architectures. Recurrent models like LSTMs, which are frequently used for temporal land cover analysis, exhibit complexity:
+
+ $$T_{LSTM} = O(e \cdot n \cdot t \cdot (h^2 + h \cdot i))$$
+ where e is epochs, t is sequence length, h is hidden dimensionality, and i is the input feature dimension. The quadratic dependence on h and the inherently sequential nature of LSTMs prevent efficient parallelization across the temporal axis, often extending training to several hours.
+
+ Spatial architectures further escalate these demands. A 2D U-Net scales linearly with the number of input channels (treating time steps as channels), but a 3D U-Net, which treats the temporal dimension as a third spatial axis, follows:
+
+ $$T_{3DUNet} = O(e \cdot n \cdot \sum_{l} k_l^3 \cdot c_{l-1} \cdot c_l \cdot s_l)$$
+
+ The cubic kernel term (k_l^3) means a $3 \times 3 \times 3$ convolution requires 27 operations per voxel, compared to 9 for a 2D equivalent. Furthermore, the memory complexity (VRAM) required to store 4D tensors (Space $\times$ Time $\times$ Channels) for large-scale mapping routinely necessitates high-end GPU clusters.
+
+ For operational deployment, LightGBM inference scales at $O(n \cdot k \cdot d)$, allowing for rapid pixel-wise mapping. By relying on CPU-only computation, this approach is not only viable for resource-constrained environments but also significantly reduces the carbon footprint of large-scale agricultural monitoring.
+
 ### Interpretation and Feature Selection

 To interpret the contributions of individual features to the model predictions, we employed SHapley Additive exPlanations (SHAP) [@shaps_2017]. This approach, based on game theory, quantifies the impact of each feature on the prediction outcome, providing insights into which features are most influential in determining land cover types.

~~-In our feature selection process, we incorporate both the mean and maximum SHAP values to comprehensively assess the influence of features on model predictions. The mean of the absolute SHAP values across all samples, provides a measure of the average impact of each feature, highlighting its overall importance across the dataset. This approach emphasizes features that consistently affect the model’s output but might underrepresented the significance of features causing substantial impacts under specific conditions. To address this, we also consider the maximum absolute SHAP values. Sorting features by their maximum absolute SHAP values allows us to identify those that have significant, albeit possibly infrequent, effects on individual predictions. This method ensures that features crucial for particular scenarios are not overlooked, thus offering a more nuanced understanding of feature importance that balances general influence with critical, situation-specific impacts.~~

+Prior to SHAP analysis, we applied a variance threshold filter to remove features with low discriminative power. Features with variance below 0.5 were excluded, as low-variance features provide minimal information for distinguishing between classes. This preprocessing step reduced the initial feature set substantially while preserving features with meaningful variation across the training samples.

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~~-Feature selection then is the union of the top 30 time series features found with both the mean and maximum SHAP values, resulting in 33 total features. This approach ensures that the selected features are both consistently influential across the dataset and capable of exerting substantial impacts under specific conditions, providing a comprehensive set of features for model training and evaluation.~~

+Feature selection then is the union of the top 30 time series features found with both the mean and maximum SHAP values, resulting in 33 total features. The choice of 30 features represents a balance between model parsimony and classification performance; preliminary experiments indicated diminishing returns beyond this threshold while smaller feature sets showed reduced accuracy. This approach ensures that the selected features are both consistently influential across the dataset and capable of exerting substantial impacts under specific conditions, providing a comprehensive set of features for model training and evaluation.

+

+For the final LightGBM model, Optuna optimized hyperparameters including the number of boosting rounds (up to 10,000 with early stopping after 200 rounds without improvement), learning rate, maximum tree depth, and regularization parameters. The early stopping mechanism prevents overfitting by halting training when validation performance plateaus, while the stratified group k-fold cross-validation ensures robust parameter selection across different data partitions (fields).

Results & Discussion

Crowd Sourced Data

~~-To address the substantial gap in available crop type datasets, particularly in developing regions, this study harnessed the power of crowdsourced data to enhance the robustness and applicability of our machine learning models. Crowdsourced data collection, an innovative approach in the agricultural domain, involves gathering data from a large number of volunteers or citizen scientists, who provide valuable ground truth information. This method has proven especially useful in areas where traditional data collection methods are challenging due to logistical, financial, or infrastructural constraints.~~

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By leveraging the YouthMappers student organization, with over 420 chapters in 80 countries, we were able to collect a large dataset of crop type observations in Tanzania. Participating YouthMappers chapters included: the Institute of Rural Development Planning - Dodoma, Institute of Rural Development Planning - Mwanza, University of Dodoma, the Nelson Mandela African Institution of Science and Technology, and the Institute of Accountancy Arusha. Moreover this exercise provided an important opportunity for students to gain practical experience in data collection, analysis, and interpretation, contributing to their professional development and capacity building in the geospatial domain.

@@ -315,7 +344,7 @@ In Figure \ref{fig:mean_shaps}, the mean SHAP values provide insights into the a

Reflecting on the colors of the bars we can see that 'B11.mean' is important in distinguishing sunflower, sorghum, and millet to a roughly equal degree, and has some small impact on distinguishing other classes. This pattern likely reflects fundamental differences in canopy structure and water content between these dryland crops and the more common maize and rice in the dataset. The SWIR bands are particularly sensitive to plant water content and canopy moisture status, as water strongly absorbs radiation at these wavelengths. Sunflower, sorghum, and millet are typically grown in drier conditions and exhibit distinct water use strategies compared to maize. Sunflower, with its deep taproot system and larger leaf area, maintains different canopy moisture levels throughout the growing season. Sorghum and millet, as drought-tolerant cereals with waxy leaf coatings and more efficient water use, exhibit characteristically different SWIR reflectance patterns than the more water-demanding maize crop. The mean SWIR reflectance over the growing season therefore captures these persistent biophysical differences in canopy water status, allowing the model to effectively separate these dryland-adapted crops from others in the training dataset. The equal importance of B11.mean across these three crops suggests that while this feature helps separate them from the broader "cereal" or "broadleaf" categories, additional features are needed to distinguish among sunflower, sorghum, and millet themselves.

~~-While 'hue.quantile.q.0.05' has the strongest effect distinguishing rice, sunflower, and to a lesser degree cotton. Measure of hue is emphasizing the overall brightness and color saturation of the crops. From the a visual inspection we can see high values of this variable are correlated with less water stressed geographies like forests and river basins. This relative water abundance might explain the correspondance of plantings of water loving rice and long tap rooted sunflowers. Looking down the list we can see that features like "EVI.standard.deviation" are most effective at isolating urban areas, as urban areas will have little variation in greeness across the year.~~
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Looking at 'B12.absolute.sum.of.changes' we can see it best differentiates shrubs, cotton, and forest. The sum of absolute changes measures the total magnitude of variation in a time series by summing the absolute differences between consecutive time steps, effectively quantifying how much a variable fluctuates over the observation period regardless of direction. When applied to the B12 band (SWIR 2100-2280nm), this metric captures the cumulative volatility in canopy water content and structural properties throughout the growing season. B12's effectiveness at distinguishing shrub, cotton, and forests using sum of absolute changes likely reflects fundamental differences in their temporal stability and management regimes. Cotton, as an intensively managed annual crop, exhibits pronounced temporal variability in the SWIR signal driven by distinct phenological transitions and management interventions. Cotton fields progress through rapid establishment after planting, vigorous vegetative growth, flowering and boll development, and then defoliation before harvest—each transition creating sharp changes in canopy water content and structure that B12 captures.

@@ -423,11 +452,15 @@ Our approach achieved robust performance with a Cohen's Kappa score of 0.82 and

The strategic use of SHAP values for feature interpretation revealed the physical and phenological mechanisms driving model predictions. For instance, we found that mean SWIR reflectance (B11.mean) effectively distinguished dryland-adapted crops (sunflower, sorghum, millet) from water-demanding crops by capturing persistent differences in canopy water status throughout the growing season. Similarly, the sum of absolute changes in B12 discriminated between cotton's managed phenological transitions and the more stable spectral signatures of natural vegetation. These insights not only enhanced model interpretability but also provided agronomically meaningful explanations that can inform future data collection and feature engineering efforts.

~~-Our study has several important limitations that suggest directions for future research. Data collection constraints impacted our results: the 2023 drought affected crop health and planting schedules, resulting in fields at varying phenological stages and some early harvests. Our concentrated data collection window (April-May 2023) captured crops primarily in late growing season, potentially missing spectral signatures from earlier phenological stages that could improve discrimination. Additionally, crop type imbalance in our dataset with underrepresentation of crops like peanuts, soybeans, and okra these crops were dropped from the study. Confusion between similar crops (e.g., cassava and maize) indicates that additional discriminative features or more extensive training data are needed for these challenging classification scenarios.~~

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Geographic generalizability remains an open question. Our model was trained exclusively on data from three districts in northern Tanzania (Arusha, Dodoma, and Mwanza), and its transferability to other regions with different agro-ecological conditions, farming practices, or crop varieties remains untested. The spectral and temporal signatures of crops can vary significantly with climate, soil conditions, and management practices, potentially limiting model performance in new geographic contexts.

~~-Finally, interpretability trade-offs persist despite our use of SHAP values: while providing valuable insights into feature importance, the large number of features (33 final features from hundreds of candidates) still presents challenges for intuitive interpretation. Future work should focus on expanding geographic coverage to assess model transferability, incorporating multi-temporal data collection throughout the growing season, developing methods to handle class imbalance, and exploring more parsimonious feature sets that balance accuracy with interpretability.~~

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+Several methodological limitations warrant consideration for future applications. Our use of monthly composites, while effective for reducing cloud contamination, may miss rapid phenological changes such as double cropping events or short-duration growth stages that occur within a single month. The linear interpolation applied to fill cloud gaps assumes gradual transitions between observations, which may not accurately capture abrupt changes in crop condition due to management interventions or stress events. Additionally, persistent cloud cover during critical phenological periods could introduce systematic biases in the time-series features. These issues might be addressed by including SAR time series data. Class boundary decisions, particularly for distinguishing between visually similar land covers such as shrub versus forest, remain challenging and depend heavily on the quality and consistency of training data. Moreover xr_fresh feature extraction, while comprehensive, may not capture all relevant phenological dynamics, suggesting that additional or alternative feature engineering approaches could further enhance model performance.

+

+Future work should focus on expanding geographic coverage to assess model transferability, incorporating multi-temporal data collection throughout the growing season, developing methods to handle class imbalance, and exploring more parsimonious feature sets that balance accuracy with interpretability.

By "turning back the clock" on deep learning, we demonstrate that carefully engineered time-series features—capturing trends, distributions, and temporal complexity—can achieve high classification accuracy without the extensive labeled datasets, computational resources, and training time required by deep learning architectures. This "lite learning" approach offers particular advantages for the resource-limited contexts that characterize much of sub-Saharan Africa and other developing regions, where both training data and computational infrastructure remain scarce.

```
diff --git a/writeup/writeup.md b/writeup/writeup.md
index 72c4f90..e03cb90 100644
```

~~--- a/writeup/writeup.md~~

+++ b/writeup/writeup.md

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Prior to SHAP analysis, we applied a variance threshold filter to remove features with low discriminative power. Features with variance below 0.5 were excluded, as low-variance features provide minimal information for distinguishing between classes. This preprocessing step reduced the initial feature set substantially while preserving features with meaningful variation across the training samples.

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Feature selection then is the union of the top 30 time series features found with both the mean and maximum SHAP values, resulting in 33 total features. The choice of 30 features represents a balance between model parsimony and classification performance; preliminary experiments indicated diminishing returns beyond this threshold while smaller feature sets showed reduced accuracy. This approach ensures that the selected features are both consistently influential across the dataset and capable of exerting substantial impacts under specific conditions, providing a comprehensive set of features for model training and evaluation.

@@ -324,7 +324,7 @@ There were a number of challenges involved with planning, and implementing a lar

Land Cover and Crop Type

~~-The distribution of primary land cover types within the training dataset used for the model are represented in Figure \ref{fig:lc_percentages}. The dataset consists of a diverse range of land cover types, each contributing differently to the total number of observations. Maize is the most prevalent land cover type, accounting for the highest percentage of the observations, followed by rice and sunflower. This is indicative of the agricultural dominance in the region being studied. Less common land covers such as millet, sorghum, and urban areas represent intermediate percentages, reflecting the heterogeneous landscape that includes both agricultural and urbanized zones. Peanuts, soybeans and okra are among the least represented land cover types in the dataset, highlighting the challenges associated with collecting sufficient training data for these categories, but also the small scale of production for these crops in the region. This figure also includes land cover types such as water, forest, shrub, and tidal areas, which are essential for providing context to the landscape but were not the focus of this study. The varied distribution of land cover types underscores the complexity of the classification task and the need for robust modeling techniques to accurately capture this diversity.~~

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\begin{figure}[H]
\centering \includegraphics[width=0.8\linewidth]
{/home/mmann1123/Documents/github/YM_TZ_crop_classifier/writeup/figures/primary_land_cover.png
} % Adjust the path and options
@@ -336,17 +336,17 @@ The distribution of primary land cover types within the training dataset used fo

The interpretation of model behavior using SHAP values has allowed for a deeper understanding of how different spectral features impact the model's predictions, which is critical for refining the feature selection process. By analyzing both the mean and maximum SHAP values, we were able to prioritize features based on their overall impact as well as their critical contributions to specific model decisions.

~~-In the two summary plots below, we display the SHAP values for each feature, to identify how much impact each feature has on the model output for pixels in the validation dataset. Features are sorted by the sum of the SHAP value across all samples. The figures bar length represents the mean contributions to explaining each predicted land class value - with different land classes represented with different colors (hues). This visualization provides a comprehensive overview of the feature importance, highlighting the key predictors that drive the model's predictions. For example, features that are highly influential for "maize" may not be as impactful for "rice" or "sorghum", reflecting the unique spectral signatures of these crops.~~
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Mean SHAP Values

In Figure \ref{fig:mean_shaps}, the mean SHAP values provide insights into the average impact of each feature across all predictions. This analysis highlights the features that consistently influence the model's output across various scenarios. For example, the mean value of B11 (B11.mean) and the 5th percentile of hue (hue.quantile.q.0.05) features were found to have substantial average impacts on model outputs, suggesting their strong relevance in distinguishing between different crop types. Reflecting on the colors of the bars we can see that 'B11.mean' is important in distinguishing sunflower, sorghum, and millet to a roughly equal degree, and has some small impact on distinguishing other classes.

~~-Reflecting on the colors of the bars we can see that 'B11.mean' is important in distinguishing sunflower, sorghum, and millet to a roughly equal degree, and has some small impact on distinguishing other classes. This pattern likely reflects fundamental differences in canopy structure and water content between these dryland crops and the more common maize and rice in the dataset. The SWIR bands are particularly sensitive to plant water content and canopy moisture status, as water strongly absorbs radiation at these wavelengths. Sunflower, sorghum, and millet are typically grown in drier conditions and exhibit distinct water use strategies compared to maize. Sunflower, with its deep taproot system and larger leaf area, maintains different canopy moisture levels throughout the growing season. Sorghum and millet, as drought-tolerant cereals with waxy leaf coatings and more efficient water use, exhibit characteristically different SWIR reflectance patterns than the more water-demanding maize crop. The mean SWIR reflectance over the growing season therefore captures these persistent biophysical differences in canopy water status, allowing the model to effectively separate these dryland-adapted crops from others in the training dataset. The equal importance of B11.mean across these three crops suggests that while this feature helps separate them from the broader "cereal" or "broadleaf" categories, additional features are needed to distinguish among sunflower, sorghum, and millet themselves.~~

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~~-While 'hue.quantile.q.0.05' has the strongest effect distinguishing rice, sunflower, and to a lesser degree cotton. Measure of hue is emphasizing the overall brightness and color saturation of the crops. From the a visual inspection we can see high values of this variable are correlated with less water stressed geographies like forests and river basins. This relative water abundance might explain the correspondence of plantings of water loving rice and long tap rooted sunflowers. Looking down the list we can see that features like "EVI.standard.deviation" are most effective at isolating urban areas, as urban areas will have little variation in greeness across the year.~~
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~~-Looking at 'B12.absolute.sum.of.changes' we can see it best differentiates shrubs, cotton, and forest. The sum of absolute changes measures the total magnitude of variation in a time series by summing the absolute differences between consecutive time steps, effectively quantifying how much a variable fluctuates over the observation period regardless of direction. When applied to the B12 band (SWIR 2100-2280nm), this metric captures the cumulative volatility in canopy water content and structural properties throughout the growing season. B12's effectiveness at distinguishing shrub, cotton, and forests using sum of absolute changes likely reflects fundamental differences in their temporal stability and management regimes. Cotton, as an intensively managed annual crop, exhibits pronounced temporal variability in the SWIR signal driven by distinct phenological transitions and management interventions. Cotton fields progress through rapid establishment after planting, vigorous vegetative growth, flowering and boll development, and then defoliation before harvest each transition creating sharp changes in canopy water content and structure that B12 captures.~~
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@@ -360,7 +360,7 @@ Looking at 'B12.absolute.sum.of.changes' we can see it best differentiates shrub

 #### Maximum SHAP Values

~~-On the other hand, Figure \ref{fig:max_shaps}, maximum SHAP values uncover features that, while perhaps not consistently influential, have high impacts under particular conditions. This aspect of the analysis can be crucial for identifying features that can cause significant shifts in model output, potentially corresponding to specific agricultural or environmental contexts. Typically we see a reshuffling of variable importance levels between maximum and mean SHAP values - reflecting differences in the average case vs edge cases. For instance, features such as "hue.median" and "B11.quantial.q.0.95" show high maximum SHAP values, indicating their pivotal roles in determining certain classes. For instance, "B11.maximum" reflects peak reflectance in the Short Wavelength Infrared (SWIR), which could be critical in identifying crops at their maximum biomass, like sunflower at full bloom compared to other crops at different stages of growth. Max SHAP values included two variables 'B12.abs.energy' and 'B12.qualtile.q.0.95' that were not included in the mean SHAP values - indicating that these features have high impacts in specific scenarios but are not consistently influential across the dataset. The appearance of these features exclusively in the maximum SHAP analysis reveals an important dimension of model behavior: the classifier has identified unusual or boundary conditions that differentiate ambiguous cases.~~
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 \begin{figure}[H]
@@ -380,7 +380,7 @@ Optuna trials tuning results selected LightGBM [@ke2017lightgbm] is a gradient b
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The classification model demonstrated robust performance across multiple land cover classes, as evidenced by the out-of-sample mean confusion matrix with a Cohen's Kappa score of 0.82 and F1-micro score of 0.85 (Table \ref{tab:metrics}), indicating substantial agreement between predicted and actual classifications. Remember that each field is treated as a ‘group’ in the group k-fold procedure to ensure that pixels from the same field are not split between the testing and training groups. The confusion matrix (Figure \ref{fig:oos_confusion}) shows high diagonal values for most classes, highlighting the model's ability to accurately identify specific land covers. For instance, rice and maize achieved out-of-sample classification accuracies of 92% and 82%, respectively. Other well-classified categories included millet, sunflower, tidal, water, shrubs, and forest, each with over 73% accuracy. However forest is primarily confused with the category shrub, which is likely a result of poor training data and the difficulty of visually determining trees versus shrubs from high-res imagery without the benefit of field visits.

~~-Categories such as sorghum, cassava and cotton displayed moderate confusion with other classes, indicating potential areas for model improvement, especially in distinguishing features that are common between similar crop types. Confusion between cassava and maize might reflect intercropping practices, where cassava is grown alongside other crops, making it difficult to isolate in the satellite imagery. The model's performance on these classes suggests that additional discriminative features or more extensive training data may be necessary to further enhance classification accuracy for these crops.~~
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 \begin{figure}[H]
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@@ -408,7 +408,7 @@ Recall (PA)   & 0.84    \\

 The overall high out-of-sample performance in Table \ref{tab:metrics} across the majority of categories suggests that the model is effective for practical applications in land cover classification, though further refinement is recommended for categories showing lower accuracy and higher misclassification rates.
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~~-We can compare our results across multiple models using the figure below in \ref{fig:model_compare} from [@kerner2024accurate]. This plot represents multiple performance metrics of land cover models that include an 'agricultural' category specifically for Tanzania. Our model's performance is indicated by the dashed line. The high level of performance - particularly for the more challenging F1 score - is not surprising given that our model is specifically trained on Tanzanian data, while the other models are typically global or regional models. On the other hand, most of these models include only a single 'agricultural' class, meaning their prediction task is a significantly easier one than the one presented here. Given this our strong out-of-sample performance is notable.~~

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 \begin{figure}[H]
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@@ -418,7 +418,7 @@ We can compare our results across multiple models using the figure below in \ref
 \end{figure}
 Source: [@kerner2024accurate]
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~~-The integration of crowdsourced data with traditional machine learning and engineered time-series features yielded a robust model for crop classification in Tanzania. While the model performed exceptionally well for crops like maize and rice, some confusion persisted among similar crop types such as sorghum and cotton. This suggests that additional discriminative features or more extensive training data may be necessary to further enhance classification accuracy for these crops. The challenges encountered, such as variability in crop cycles and challenges of crop identification, highlight the complexities of agricultural monitoring in resource-limited settings. Addressing these issues in future research could improve model performance and generalizability. Overall, our findings demonstrate the practicality of using efficient, interpretable machine learning methods in conjunction with community-driven data collection to advance agricultural monitoring in developing regions.~~

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 ### Land Cover Product

@@ -452,13 +452,13 @@ Our approach achieved robust performance with a Cohen's Kappa score of 0.82 and

 The strategic use of SHAP values for feature interpretation revealed the physical and phenological mechanisms driving model predictions. For instance, we found that mean SWIR reflectance (B11.mean) effectively distinguished dryland-adapted crops (sunflower, sorghum, millet) from water-demanding crops by capturing persistent differences in canopy water status throughout the growing season. Similarly, the sum of absolute changes in B12 discriminated between cotton's managed phenological transitions and the more stable spectral signatures of natural vegetation. These insights not only enhanced model interpretability but also provided agronomically meaningful explanations that can inform future data collection and feature engineering efforts.
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~~-Our study has several important limitations that suggest directions for future research. Data collection constraints impacted our results: the 2023 drought affected crop health and planting schedules, resulting in fields at varying phenological stages and some early harvests. Our concentrated data collection window (April-May 2023) captured crops primarily in late growing season, potentially missing spectral signatures from earlier phenological stages that could improve discrimination. Additionally, crop type imbalance in our dataset with under representation of crops like peanuts, soybeans, and okra these crops were dropped from the study. Confusion between similar crops (e.g., cassava and maize) indicates that additional discriminative features or more extensive training data are needed for these challenging classification scenarios.~~
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Geographic generalizability remains an open question. Our model was trained exclusively on data from three districts in northern Tanzania (Arusha, Dodoma, and Mwanza), and its transferability to other regions with different agro-ecological conditions, farming practices, or crop varieties remains untested. The spectral and temporal signatures of crops can vary significantly with climate, soil conditions, and management practices, potentially limiting model performance in new geographic contexts.

Finally, interpretability trade-offs persist despite our use of SHAP values: while providing valuable insights into feature importance, the large number of features (33 final features from hundreds of candidates) still presents challenges for intuitive interpretation.

~~-Several methodological limitations warrant consideration for future applications. Our use of monthly composites, while effective for reducing cloud contamination, may miss rapid phenological changes such as double cropping events or short-duration growth stages that occur within a single month. The linear interpolation applied to fill cloud gaps assumes gradual transitions between observations, which may not accurately capture abrupt changes in crop condition due to management interventions or stress events. Additionally, persistent cloud cover during critical phenological periods could introduce systematic biases in the time-series features. These issues might be addressed by including SAR time series data. Class boundary decisions, particularly for distinguishing between visually similar land covers such as shrub versus forest, remain challenging and depend heavily on the quality and consistency of training data. Moreover xr_fresh feature extraction, while comprehensive, may not capture all relevant phenological dynamics, suggesting that additional or alternative feature engineering approaches could further enhance model performance.~~
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Future work should focus on expanding geographic coverage to assess model transferability, incorporating multi-temporal data collection throughout the growing season, developing methods to handle class imbalance, and exploring more parsimonious feature sets that balance accuracy with interpretability.